

A non-inverting amplifier (R_1 is input resistor, R_2 is feedback resistor) is built with a gain bandwidth product of 10 MHz and a non-zero output resistance $r_o = 75 \Omega$ (assume $r_d = \infty$).

What is the maximum purely capacitive load, C_L , that could be added to the output such that $\phi_M \geq 45^\circ$ for:

$$GBP = f_t = 10 \text{ MHz}, r_o = 75, \phi_M = 45$$

$$V_n = V_{op} * R_1 / (R_1 + R_2)$$

$$V_{op} = V_{out} * ((R_1 + R_2) || C) / ((R_1 + R_2) || C + r_o)$$

$$((R_1 + R_2) * 1 / (j\omega C)) / ((R_1 + R_2) + 1 / (j\omega C)) / ((R_1 + R_2) * 1 / (j\omega C)) / ((R_1 + R_2) + 1 / (j\omega C)) + r_o$$

$$((R_1 + R_2) / ((j\omega C)(R_1 + R_2) + 1)) / ((R_1 + R_2) / ((j\omega C)(R_1 + R_2) + 1)) + r_o$$

$$(R_1 + R_2) / ((R_1 + R_2) + r_o(j\omega C)(R_1 + R_2) + 1)$$

$$1 / (1 + r_o(j\omega C) + 1 / (R_1 + R_2))$$

$$1 / (1 + r_o(j\omega C) + r_o / (R_1 + R_2))$$

$$R_1 + R_2 \gg 20k \gg r_o = 75$$

$$V_{op} = V_{out} / (1 + r_o(j\omega C))$$

$$V_n = R_1 / (R_1 + R_2) * V_{out} / (1 + r_o(j\omega C))$$

$$V_n = V_{out} * (R_1 / (R_1 + R_2)) / (1 + r_o(j\omega C))$$

$$|B| = V_n / V_{out} = (R_1 / (R_1 + R_2)) / (1 + r_o(j\omega C))$$

$$|B| = (R_1 / (R_1 + R_2)) / (1 + jf * 2\pi r_o C) = B_0 / (1 + jf / f_o)$$

$$B_0 = (R_1 / (R_1 + R_2)), f_o = 1 / (2\pi r_o C)$$

$$a_0 * f_b = f_t, a_0 = f_t / f_b = 10 \text{ M} / f_b$$

$$T(jf) = (R_1 / (R_1 + R_2)) (1 / (1 + jf / f_o)) ((f_t / f_b) / (1 + jf / f_b))$$

$$\angle T(jf) = -\arctan(f / f_b) - \arctan(f / f_o) = -135$$

$$|T(jf)| = (R_1 / (R_1 + R_2)) (f_t / f_b) / (\sqrt{1 + f^2 / f_o^2} * \sqrt{1 + f^2 / f_b^2}) = 1$$

solve for f and f_o in terms of f_b

$$\angle T(jf) = -\arctan(f / f_b) - \arctan(f / f_o) = -135$$

$$f \gg f_b$$

$$-90 - \arctan(f / f_o) = -135$$

$$\arctan(f / f_o) = 45$$

$$f = f_o$$

$$f \gg f_b$$

$$|T(jf)| = (R_1 / (R_1 + R_2)) (f_t / f_b) / (\sqrt{1 + f^2 / f_o^2} * f / f_b) = (R_1 / (R_1 + R_2)) (f_t) / (\sqrt{2} * f) = 1$$

$$f = (R_1 / (R_1 + R_2)) (f_t) / \sqrt{2} = f_o = 1 / (2\pi r_o C) = 3.5355 \text{ MHz}$$

$$C = 1 / (2\pi r_o f_t R_1 / ((R_1 + R_2) \sqrt{2})) = (R_1 + R_2) \sqrt{2} / (2\pi r_o f_t R_1)$$

$$(a) R_1 = R_2 = 20 \text{ k}\Omega,$$

$$C = (20k + 20k) \sqrt{2} / (2\pi 75 10 \text{ M} 20k) = 600.211 \text{ pF}$$

$$(b) R_1 = R_2 = 200 \text{ k}\Omega,$$

$$C = (200k + 200k) \sqrt{2} / (2\pi 75 10 \text{ M} 200k) = 600.211 \text{ pF}$$

$$(c) R_1 = 2 \text{ k}\Omega, R_2 = 18 \text{ k}\Omega, \text{ and}$$

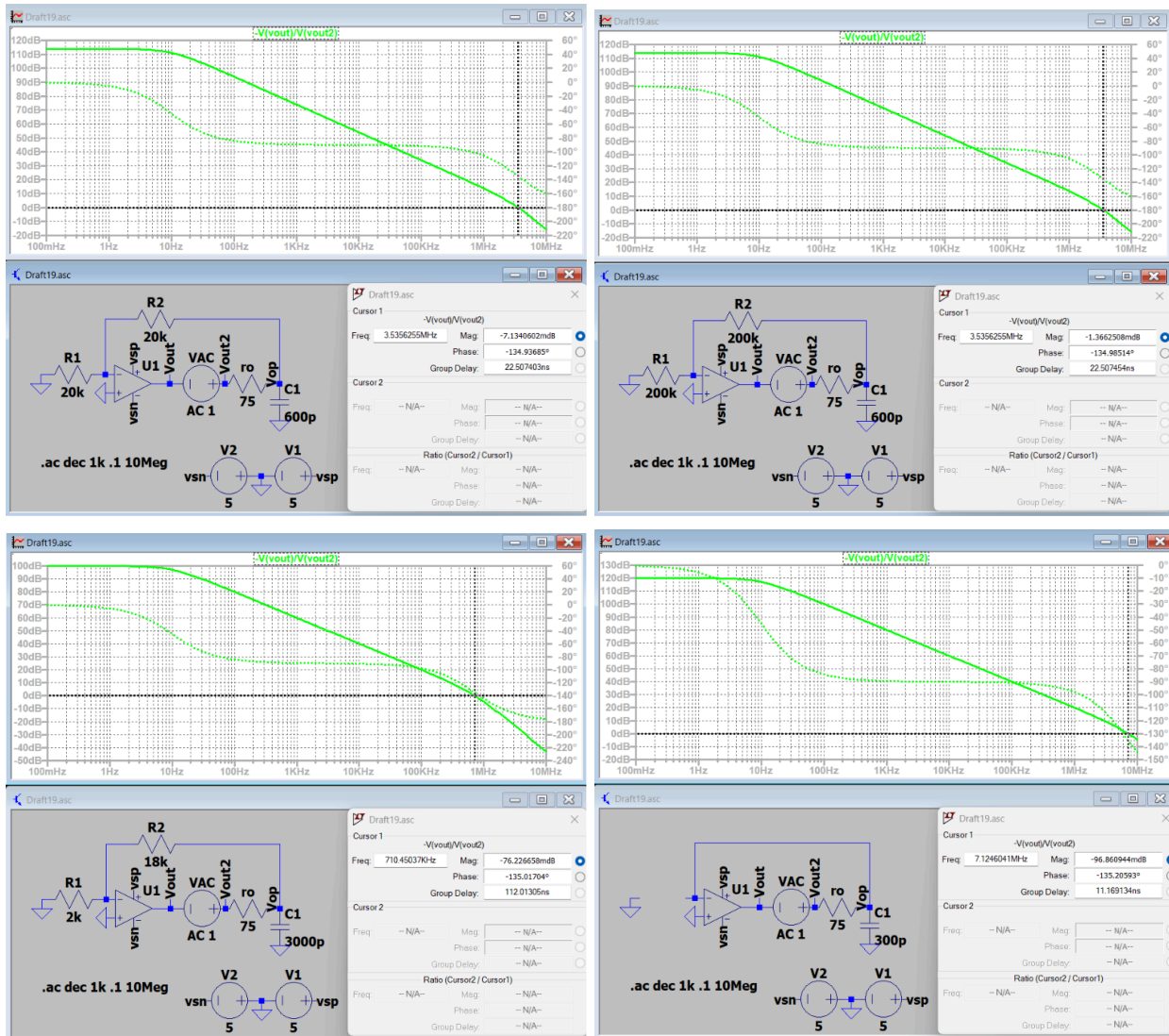
$$C = (2k + 18k) \sqrt{2} / (2\pi 75 10 \text{ M} 2k) = 3001.054 \text{ pF}$$

$$(d) R_1 = \infty, R_2 = 0 \Omega \text{ (voltage follower)}$$

$$C = (\infty + 0) \sqrt{2} / (2\pi 75 10 \text{ M} \infty) = \sqrt{2} / (2\pi 75 10 \text{ M}) = 300.105 \text{ pF}$$

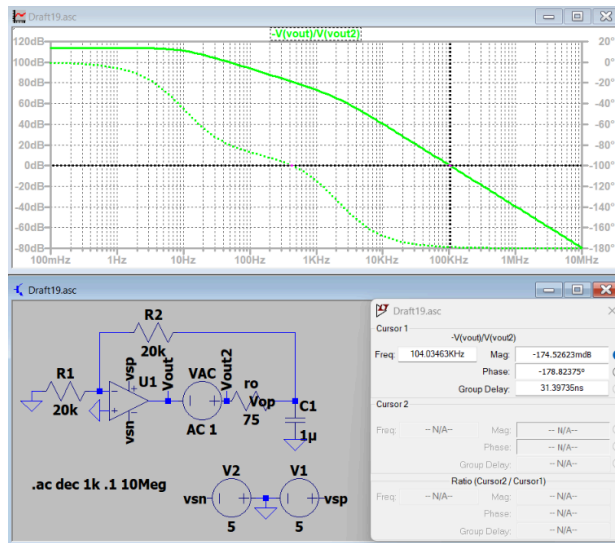
P2)

Verify your answers to question 1 using LTspice: simulate and measure $T(jf)$ for each part. It is suggested to use UniversalOpAmp1 and set the GBP accordingly. Note that this model does not include r_o ; therefore, an external resistor must be added to model the output resistance.



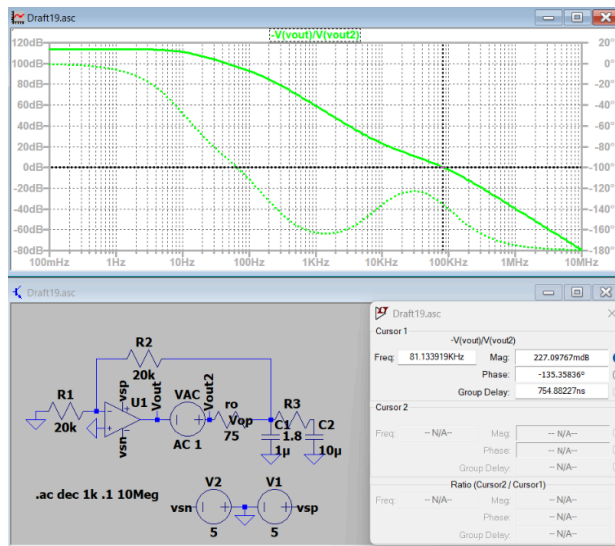
P3)

In LTspice: using the same circuit in question 1 and $R1 = R2 = 20\text{ k}\Omega$, let $C_L = 1\text{ }\mu\text{F}$, resulting in a very small ϕ_M (note, since there are only two poles in this circuit, it will technically never go unstable). Modify the circuit such that $\phi_M \approx 45^\circ$ by:



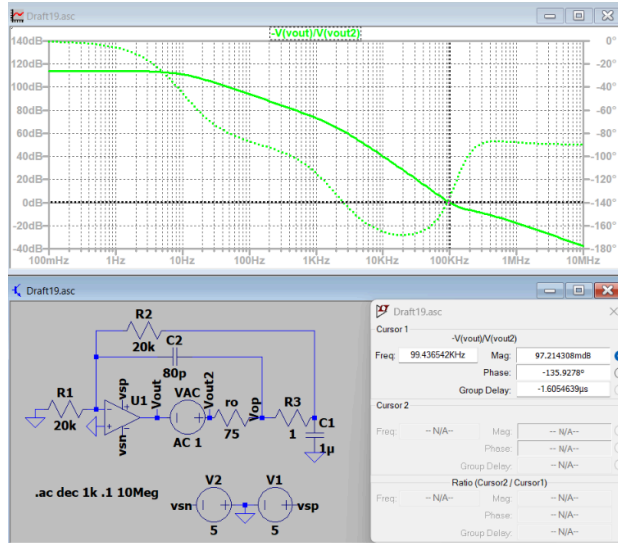
$\phi_M \approx 1.2^\circ$

(a) Implementing a snubber network (resistor+capacitor in parallel with C_L) and tuning as described the lecture 17 notes, and



$\phi_M = 45^\circ$

(b) Adding a feedback capacitor, C_f , and output series resistor, R_s , (also shown in lecture 17 notes). Assume that $R_s = 1\text{ }\Omega$.



Comment on the expected low-frequency gain error and gain peaking and/or oscillatory behavior of each system. Note that this can be done by simply interpreting the simulated $T(jf)$ but could also be simulated via a transient analysis with step input (akin to default design project input).

None of these significantly affect A_{ideal} at DC. The initial system with $C_L=600\text{pF}$ has a phase margin of about 45° , slightly underdamped, slight gain peaking and overshoot. Increasing to 1μ gives a very small phase margin, with significant gain peaking, still technically stable because only 2 poles. The snubber reduces peaking by dissipating energy in the load capacitance, and giving a more damped response. The feedback capacitor and output series resistor reduces overshoot with lead compensation and isolating the load.

P4)

-6050 Only: Book Problem 8.50 (a+b only): This is more a qualitative problem than a quantitative one. Assume a generic op amp (such as the 741 amps: GBP = 1 MHz, fb = 5 Hz, and ao = 200 V/mV). You may also assume either:

- that the crossover frequency, fx, occurs at (1) a higher frequency than any/all poles and zeros within the feedback network and (2) fx > fb;
- OR, use component values: – Fig. 3.11: R1 = R2 = 10 kΩ, C1 = 1 μF, C2 = 10 nF. – Fig. 3.32: R1 = 1 kΩ, R2 = 10 kΩ, R3 = 100 kΩ, C1 = 100 nF, C2 = 10 nF

8.50: Assuming constant-GBP op amps, use linearized Bode plots to investigate the stability of (a) the wideband band-pass filter of Fig. 3.11, (b) the multiple-feedback low-pass filter of Fig. 3.32, and (ignore)(c) the -KRC low-pass filter of Problem 3.32

a)

The last two circuits can be merged as in Fig. 3.11a to give a *band-pass* response. Letting $Z_1 = (R_1 C_1 s + 1)/C_1 s$ and $Z_2 = R_2/(R_2 C_2 s + 1)$, we get $H(s) = -Z_2/Z_1$, or

$$H(s) = -\frac{R_2}{R_1} \frac{R_1 C_1 s}{R_1 C_1 s + 1} \frac{1}{R_2 C_2 s + 1} \quad (3.28)$$

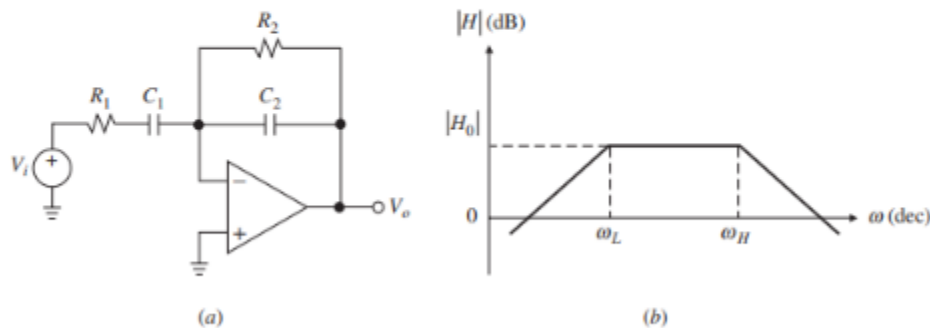


FIGURE 3.11
Wideband band-pass filter.

b)

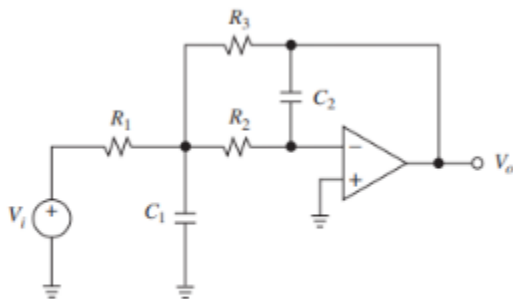
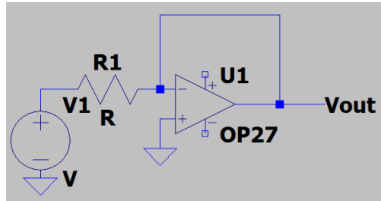


FIGURE 3.32
Multiple-feedback low-pass filter.

Using assumption 1 fx > feedback poles and fx > fb

a)

fx > fp, fb, caps are shorts



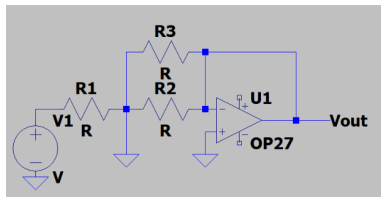
$$B=1$$

$$\text{ROC} = 0 - -20 = 20$$

$$\phi_M=90$$

stable

b)



same $B=1$, $\text{ROC} = 0 - -20 = 20$, $\phi_M=90$, stable

P5) Book Problem 8.55.

In Fig. P8.55 let $R_1 = R_2 = R_4 = 100 \text{ k}$, $R_3 = 10 \text{ k}$, and let the op amp have $a_0 = 10^5 \text{ V/V}$, $f_1 = 10 \text{ kHz}$, $f_2 = 200 \text{ kHz}$, and $f_3 = 2 \text{ MHz}$. (a) Verify that the circuit is unstable. (b) Use input-lag compensation to stabilize it for $\phi_m = 45^\circ$. (c) Find the closed-loop bandwidth after compensation.

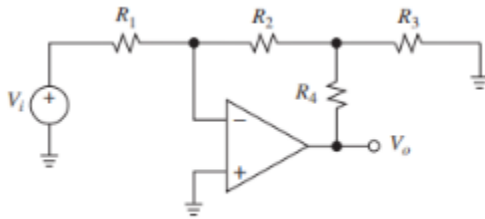


FIGURE P8.55

a)

$$V_x = V_{out} \left(\frac{(R_1 + R_2) \parallel R_3}{((R_1 + R_2) \parallel R_3) + R_4} \right)$$

$$V_n = V_x \frac{R_1}{(R_1 + R_2)} = V_{out} \frac{R_1}{((R_1 + R_2) \parallel R_3) + R_4} = V_{out} \cdot 0.043478$$

$$1/B = 23$$

$$T = 100k(1/23) / ((1 + jf/10k)(1 + jf/200k)(1 + jf/2M))$$

$$|T(f_x)| = 1 = 100k(1/23) / ((1 + jf_x/10k)(1 + jf_x/200k)(1 + jf_x/2M))$$

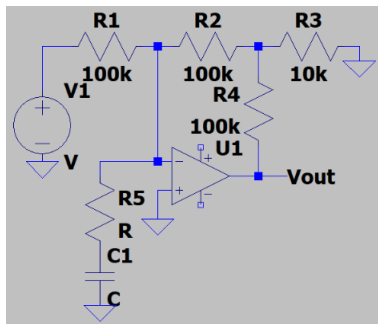
$$f_x = 2.3653 \text{ MHz}$$

$$\angle T(f_x) = -\arctan(f_x/10k) - \arctan(f_x/200k) - \arctan(f_x/2M) = -260.1^\circ$$

$$\phi_m = -80.1^\circ$$

unstable

b)



let $Z = R_5 + C_1$

$$V_x = V_{out} \left(\frac{(R_1 \parallel Z) + R_2}{((R_1 \parallel Z) + R_2) \parallel R_3} \right) \parallel R_4$$

$$V_n = V_x \frac{R_1 \parallel Z}{(R_1 \parallel Z) + R_2}$$

$$V_n = V_{out} \left(\frac{(R_1 \parallel Z) + R_2}{((R_1 \parallel Z) + R_2) \parallel R_3} \right) \parallel R_4 \cdot \frac{R_1 \parallel Z}{((R_1 \parallel Z) + R_2)}$$

$$V_n = V_{out} \left(\frac{((R_1 \parallel Z) + R_2) \parallel R_3}{((R_1 \parallel Z) + R_2) \parallel R_3 + R_4} \right) \cdot \frac{R_1 \parallel Z}{((R_1 \parallel Z) + R_2)}$$

$$V_n = V_{out} \left(\frac{R_1 \parallel Z}{((R_1 \parallel Z) + R_2) \parallel R_3 + R_4} \right)$$

$$\left(\frac{1}{((R_1 \parallel Z) + R_2) \parallel R_3 + R_4} \right) \cdot \frac{R_1 \parallel Z}{((R_1 \parallel Z) + R_2)}$$

$$V_n = V_{out} \frac{Z}{(23Z + 1.2M)} = \frac{1}{(23 + 1.2M/Z)}$$

$$1/B = 23 + 1.2M/Z$$

$$|a(f_2)| = 100k / ((1 + jf_2/10k)(1 + jf_2/200k)(1 + jf_2/2M)) = 3513.598$$

$$1/B = 23 + 1.2M/Z = 3513$$

$$Z = 343.78$$

$$f_p = 1/(2\pi R C)$$

$$R = 343.78$$

$$C = 23\text{nF}$$

c)

$$f_B \sim f_2 = 200\text{k}$$